

Day 2

IoT and Edge AI

Hands-on session

BUSINESS MODEL CANVAS

KEY PARTNERS Who are your key partners?	KEY ACTIVITIES What are the activities you perform every day to deliver your value proposition?	VALUE PROPOSITION What is the value you deliver to your customer? What is the customer need that your value proposition addresses?	CUSTOMER RELATIONSHIPS What relationship does each customer segment expect you to establish and maintain?	CUSTOMER SEGMENTS Who are your customers?
	KEY RESOURCES What are the resources you need to deliver your value proposition?		CHANNELS How do your customer segments want to be reached?	
COST STRUCTURE What are the important costs you make to deliver the value proposition?			REVENUE STREAMS How do customers reward you for the value you provide to them?	



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 171 Second Street, Suite 300, San Francisco, California, 94105, USA.



My classes (4)

● ~~Business models and the Internet of Things.~~

● ~~Business Model Canvas~~

—

● IoT / Edge AI project (hands-on)

—

● The IoT Stack - How to build your IoT project, digital transformation and more.

● IoT Model Canvas

—

● Scale your IoT / Edge AI projects (hands-on)

Today

Today

- Buy VS Build
- Let's build together a simple Edge AI object detection project with Raspberry Pi 4 and a Webcam

First question...

Buy

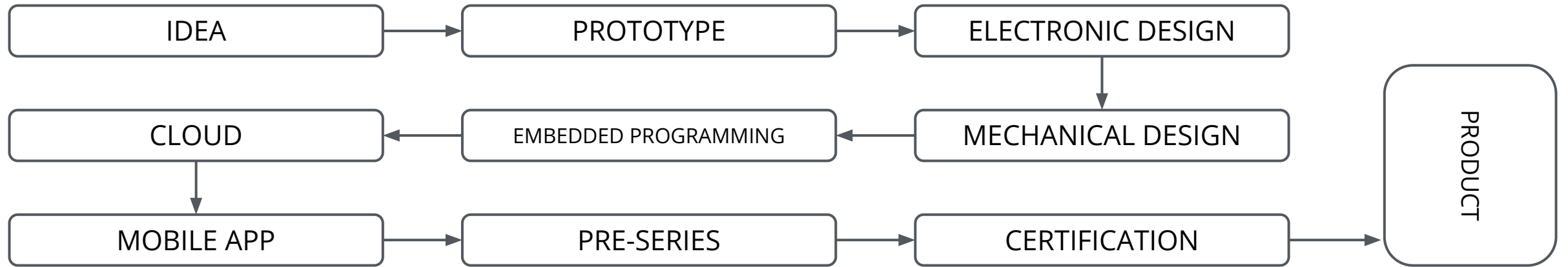
VS

Build



What do you need to build an IoT Solution?

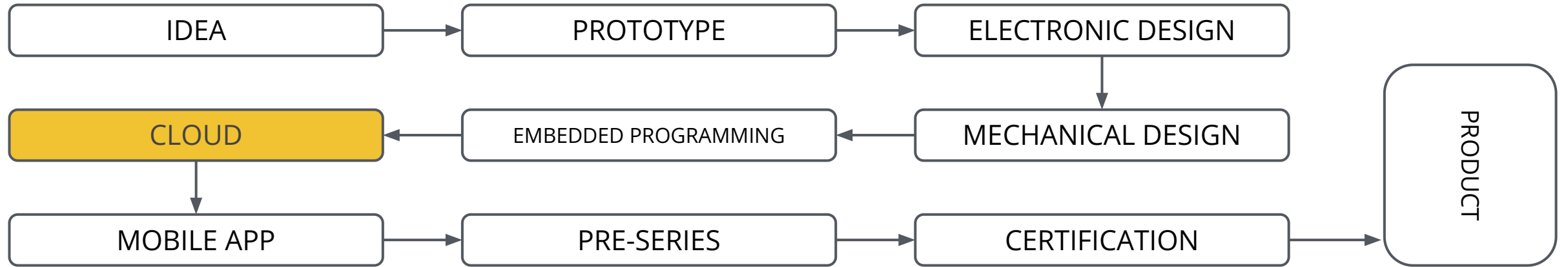
What do you need to build an IoT Solution?



“Product cycle” inspired by Elaine Chen, *Bringing a Hardware Product to Market* and Bruno Cendón

What do you need to build an IoT Solution?

How the Cloud affect other parts of the solution?



"Product cycle" inspired by Elaine Chen, *Bringing a Hardware Product to Market* and Bruno Cendón

The Value Chain of the Internet of Things

The Value Chain of the Internet of Things



BUSINESS MODEL

The Value Chain of the Internet of Things



The Value Chain of the Internet of Things



Understand the market, needs of your project and capacities of your team.

What are the key vendors in the market and how they can help you to succeed?

The Value Chain of the Internet of Things



The Value Chain of the Internet of Things



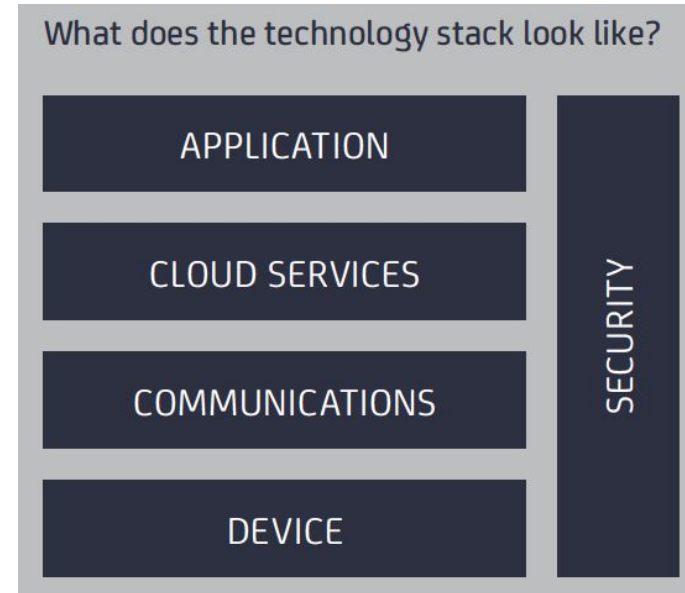
Make a small amount of units and try them in a real environment! Sell them!

Take into account security :)

The Value Chain of the Internet of Things



The Value Chain of the Internet of Things



The Value Chain of the Internet of Things



The Value Chain of the Internet of Things



Once validated the pilots and business models, this is a moment to start manufacturing the solution and sell to more clients, regions, countries, etc.

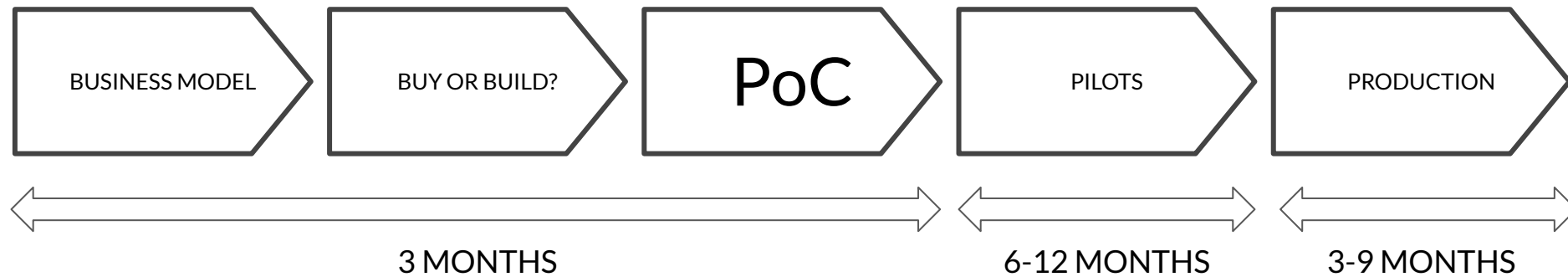
COMMON ERRORS

The system designed for pilots does not scale on production!

The Value Chain of the Internet of Things



The Value Chain of the Internet of Things



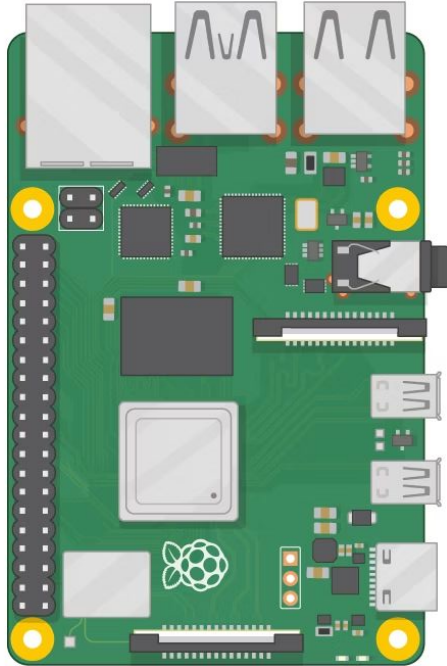
The Internet of Things...



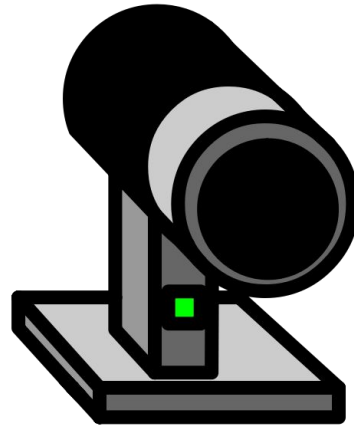
What's next?

Let's build together a simple Edge AI
object detection project with
Raspberry Pi 4 and a Webcam

Edge AI Project: Object Detection



Raspberry Pi 4



USB Webcam with
microphone



Micro SD card

Create Edge Impulse account

<https://edgeimp.com/master25>



Raspberry Pi 4 - Edge Impulse

docs.edgeimpulse.com/hardware/boards/raspberry-pi-4

EDGE IMPULSE

Knowledge Studio Hardware Tools APIs Tutorials Projects Datasets

Seeed SenseCAP A1101
macOS devices
Linux x86_64 devices
Mobile phone

BOARDS

MCU
MCU + AI accelerator
CPU
Arduino UNO Q
Microchip SAMA7G54
Raspberry Pi 4
Raspberry Pi 5
Renesas RZ/G2L
Texas Instruments SK-AM62
CPU + AI accelerator
GPU

DEPLOYMENTS

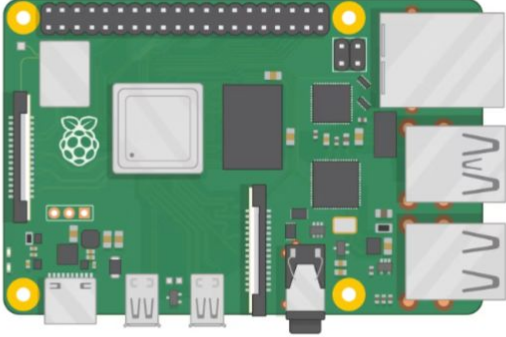
Overview
C++ library
Arduino
DRP-AI library
Zephyr
WebAssembly library
Arm Keil MDK CMSIS-Pack

BOARDS > CPU

Raspberry Pi 4

The Raspberry Pi 4 is a versatile Linux development board with a quad-core processor running at 1.5GHz, a GPIO header to connect sensors, and the ability to easily add an external microphone or camera - and it's fully supported by Edge Impulse. You'll be able to sample raw data, build models, and deploy trained machine learning models directly from the Studio.

In addition to the Raspberry Pi 4 we recommend that you also add a camera and / or a microphone. Most popular USB webcams and the [Camera Module](#) work fine on the development board out of the box.



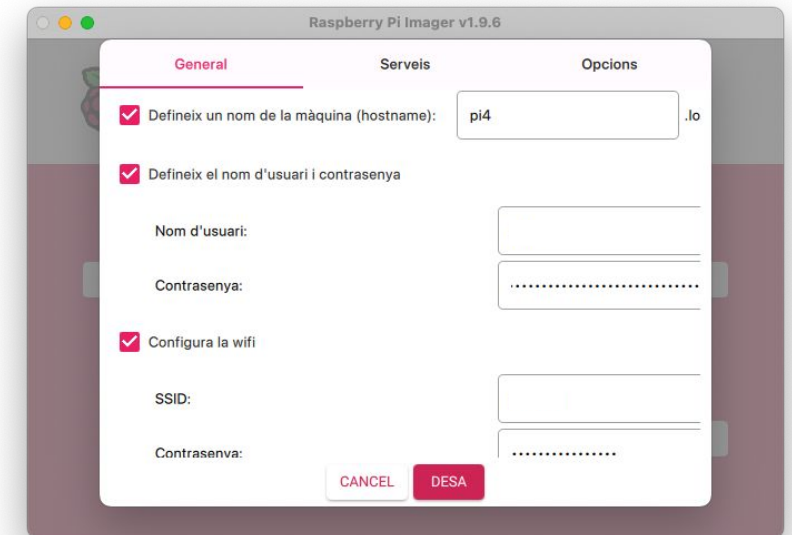
Raspberry Pi 4

On this page

Prerequisites

- Headless Setup
- Setup with a Screen
- Installing dependencies
- Install with Docker
- Connecting to Edge Impulse
- Next steps: building a machine learning model
- Deploying back to device
- Image model?
- Troubleshooting
- Wrong OS bits

<https://docs.edgeimpulse.com/hardware/boards/raspberry-pi-4>



<https://www.raspberrypi.com/software/>

arp -na | grep -i dc:a6:32

```
marcpous@Marc-Pous-GJQJX9NM7X ~ % arp -na | grep -i dc:a6:32  
? (192.168.1.136) at dc:a6:32:b1:7:43 on en0 ifscope [ethernet]
```

```
marcpous@Marc-Pous-GJQJX9NM7X ~ % ssh marc@192.168.1.136  
marc@192.168.1.136's password:  
Linux pi4 6.12.47+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.47-1+rpt1 (2025-09-16) aarch64
```

```
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.
```

```
Last login: Thu Nov 13 10:11:38 2025 from 192.168.1.145
```

```
marc@pi4:~ $
```

```
sudo apt update
```

```
curl -sL https://deb.nodesource.com/setup_20.x | sudo bash -
```

```
sudo apt install -y gcc g++ make build-essential nodejs sox
```

```
gststreamer1.0-tools gststreamer1.0-plugins-good
```

```
gststreamer1.0-plugins-base gststreamer1.0-plugins-base-apps
```

```
sudo npm install edge-impulse-linux -g --unsafe-perm
```



```
marc@pi4:~ $ edge-impulse-linux
Edge Impulse Linux client v1.18.2
```

```
? Select a microphone (Use arrow keys)
> vc4-hdmi - vc4-hdmi-0
  vc4-hdmi - vc4-hdmi-1
  bcm2835_headpho - bcm2835 Headphones
  USB-Audio - Logitech BRIO
```

```
marc@pi4:~ $ edge-impulse-linux
Edge Impulse Linux client v1.18.2
```

```
? Select a microphone USB-Audio - Logitech BRIO
[SER] Using microphone hw:3,0
[GST] checking for /etc/os-release
? Select a camera (Use arrow keys)
> Logitech BRIO (/dev/video0)
  Logitech BRIO (/dev/video2)
```

```
marc@pi4:~ $ edge-impulse-linux
Edge Impulse Linux client v1.18.2
```

```
? Select a microphone USB-Audio - Logitech BRIO
[SER] Using microphone hw:3,0
[GST] checking for /etc/os-release
? Select a camera Logitech BRIO (/dev/video0)
[SER] Connected to camera Logitech BRIO (/dev/video0)
[WS ] Connecting to wss://remote-mgmt.edgeimpulse.com
[WS ] Connected to wss://remote-mgmt.edgeimpulse.com
? What name do you want to give this device? Pi 4
[WS ] Device "Pi 4" is now connected to project "Object Detection - Rubber Duckies". To connect to another project, run `edge-impulse-linux --clean`.
[WS ] Go to https://studio.edgeimpulse.com/studio/783798/acquisition/training to build your machine learning model!
```


Object Detection - Rubber Duckies

studio.edgeimpulse.com/studio/783798/acquisition/training?page=1

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EDGE IMPULSE

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EON Tuner

Impulse design

Create impulse

Image

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GETTING STARTED

Marc using Arduino UNO Q / Object Detection - Rubber Duckies

ENTERPRISE

Target: Raspberry Pi 5

Dataset

Data sources

Synthetic data

Labeling queue (0)

AI labeling

DATA COLLECTED

212 items

TRAIN / TEST SPLIT

82% / 18%

Dataset

Training (168)

Test (41)

Post-processing (3)

unknown.67qkjj9n

unknown.67qkf6ai

unknown.67qkera4

unknown.67qkefq9

Captura de pantalla 20:

Picture.64rn5jhc

Picture.64rn6l0s

Picture.64rn6suk

Picture.64rn77un

Collect data

Device

Pi 4

Name

sample

Sensor

Camera (640x480)

Start sampling

Camera feed

Click on a sample to load...

Collect data



Device ?

Pi 4



Name

sample

Sensor

Camera (640x480)



Camera feed



Start sampling

```
marc@pi4:~ $ edge-impulse-linux
Edge Impulse Linux client v1.18.2

? Select a microphone USB-Audio - Logitech BRIO
[SER] Using microphone hw:3,0
[GST] checking for /etc/os-release
? Select a camera Logitech BRIO (/dev/video0)
[SER] Connected to camera Logitech BRIO (/dev/video0)
[WS ] Connecting to wss://remote-mgmt.edgeimpulse.com
[WS ] Connected to wss://remote-mgmt.edgeimpulse.com
? What name do you want to give this device? Pi 4
[WS ] Device "Pi 4" is now connected to project "Object Detection - Rubber Duckies". To connect to another project, run `edge-impulse-linux --clean`.
[WS ] Go to https://studio.edgeimpulse.com/studio/783798/acquisition/training to build your machine learning model!
[WS ] Incoming sampling request {
  path: '/api/training/data',
  label: 'sample',
  length: 60000,
  interval: 0,
  hmacKey: '81219f1e54840487892341f8689cbcba',
  sensor: 'Camera (640x480)'
}
[SER] Uploading sample to https://ingestion.edgeimpulse.com/api/training/data...
[SER] Sampling finished
[WS ] Incoming sampling request {
  path: '/api/training/data',
  label: 'sample',
  length: 60000,
  interval: 0,
  hmacKey: '81219f1e54840487892341f8689cbcba',
  sensor: 'Camera (640x480)'
}
[SER] Uploading sample to https://ingestion.edgeimpulse.com/api/training/data...
[SER] Sampling finished
```

Object Detection - Rubber Duck

studio.edgeimpulse.com/studio/783798/label-object-detection

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https://studio.edgeimpulse.com/studio/783798/label-object-detection

Marc using Arduino UNO Q / Object Detection - Rubber Duckies

ENTERPRISE

Target: Raspberry Pi 5

Dataset

Data sources

Synthetic data

Labeling queue (2)

AI labeling

Tired of labeling by hand? Try out AI labeling - our new LLM-powered labeling tool, available for everyone.

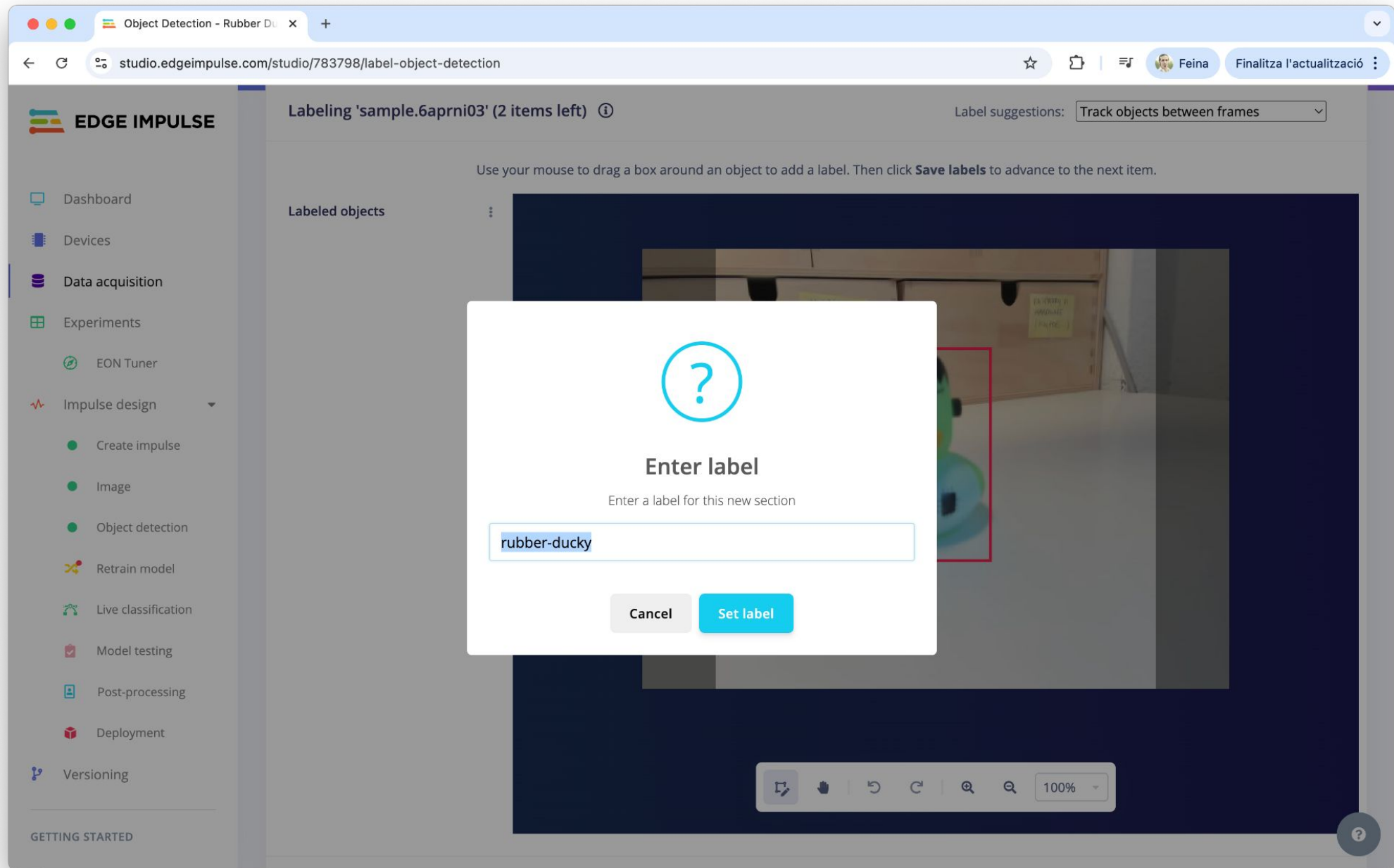
Labeling 'sample.6aprn03' (2 items left)

Label suggestions: Track objects between frames

Use your mouse to drag a box around an object to add a label. Then click **Save labels** to advance to the next item.

Labeled objects

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Object Detection - Rubber D

studio.edgeimpulse.com/studio/783798/impulse/1/create-impulse

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Marc using Arduino UNO Q / Object Detection - Rubber Duckies

ENTERPRISE

Target: Raspberry Pi 5

Impulse #1

An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.

Image data

Input axes

image

Image width96Image height96

Resize modeFit shortestest axis

Image

NameImage

Input axes (1)Imageimage

Object Detection (Images)

NameObject detection

Input featuresImage

Output features1 (rubber-ducky)

Output features

1 (rubber-ducky)

Add a processing block

Add a learning block

Save Impulse

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Object Detection - Rubber D

studio.edgeimpulse.com/studio/783798/impulse/1/dsp/image/4

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Marc using Arduino UNO Q / Object Detection - Rubber Ducks

ENTERPRISE

Target: Raspberry Pi 5

Parameters

Generate features

Raw data

0xafb1a8, 0x969894, 0x8b8d8b, 0x8d8d8b, 0x8d8d8b, 0x8c8c8a, 0x8a8a8a, 0x878685, 0x888883, 0x8d8a86, 0...

0xafb1a8, 0x969894, 0x8b8d8b, 0x8d8d8b, 0x8d8d8b, 0x8c8c8a, 0x8a8a8a, 0x878685, 0x888883, 0x8d8a86, 0...

Parameters

Image

Color depth ②

RGB

Save parameters

DSP result

Image

Processed features

0.6863, 0.6941, 0.6588, 0.5882, 0.5961, 0.5804, 0.5451, 0.5529, 0.5451, 0.5529, 0.5529, 0.5451, 0.552...

On-device performance ②

PROCESSING TIME

1 ms.

PEAK RAM USAGE

4 KB

Object Detection - Rubber D

studio.edgeimpulse.com/studio/783798/impulse/1/dsp/image/4/generate-features

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Marc using Arduino UNO Q / Object Detection - Rubber Duckies

ENTERPRISE

Target: Raspberry Pi 5

Parameters

Generate features

Training set

Data in training set

170 items

Classes

1 (rubber-ducky)

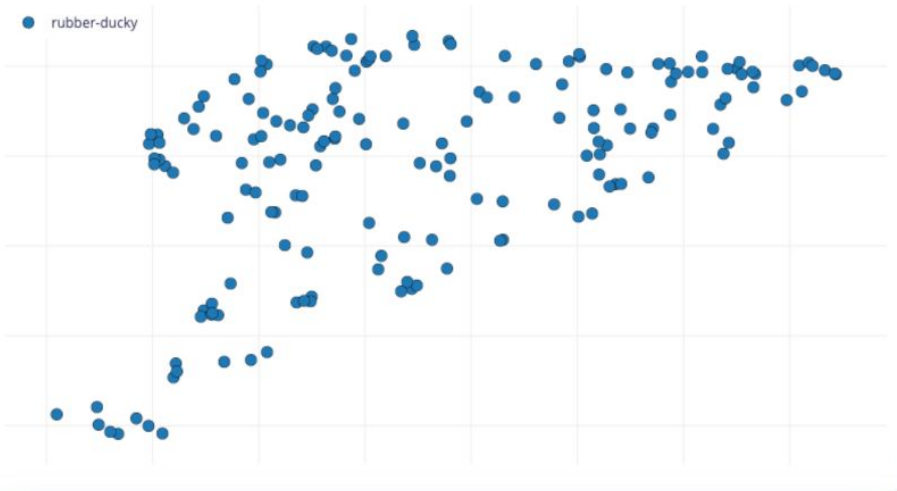
Generate features

Feature generation output

(0)

Feature explorer

rubber-ducky



On-device performance

PROCESSING TIME

1 ms.

PEAK RAM USAGE

4 KB

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?

https://studio.edgeimpulse.com/studio/783798/impulse/1/dsp/image/4/generate-features

Object Detection - Rubber D

studio.edgeimpulse.com/studio/783798/impulse/1/learning/keras-object-detection/5

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Marc using Arduino UNO Q / Object Detection - Rubber Duckies

ENTERPRISE

Target: Raspberry Pi 5

Neural Network settings

Training settings

Number of training cycles 40

Use learned optimizer

Learning rate 0.001

Training processor GPU

Data augmentation

Advanced training settings

Neural network architecture

Input layer (27,648 features)

FOMO (Faster Objects, More Objects) MobileNetV2 0.35

Choose a different model

Output layer (1 classes)

Save & train

Training output

Model

Model version: Unoptimized (float32)

Last training performance (validation set)

F1 SCORE 85.0%

Confusion matrix (validation set)

	BACKGROUND	RUBBER-DUCKY
BACKGROUND	100.0%	0.0%
RUBBER-DUCKY	12.9%	87.1%
F1 SCORE	1.00	0.85

Metrics (validation set)

METRIC	VALUE
Precision (non-background)	0.83
Recall (non-background)	0.87
F1 Score (non-background)	0.85

On-device performance

Engine: EON™ Compiler (RAM optimized)

INFERRING TIME 1 ms.

PEAK RAM USAGE 435.2K

FLASH USAGE 112.4K

Object Detection - Rubber Duckies

studio.edgeimpulse.com/studio/783798/impulse/1/deployment

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Marc using Arduino UNO Q / Object Detection - Rubber Duckies

ENTERPRISE

Target: Raspberry Pi 5

Configure your deployment

You can deploy your impulse to any device. This makes the model run without an internet connection, minimizes latency, and runs with minimal power consumption. [Read more.](#)

Search deployment options

DEFAULT DEPLOYMENT

C++ library

A portable C++ library with no external dependencies, which can be compiled with any modern C++ compiler.

MODEL OPTIMIZATIONS

Model optimizations can increase on-device performance but may reduce accuracy.

EON™ Compiler

Same accuracy, 19% less RAM.

Quantized (int8)

Selected ✓

	IMAGE	OBJECT DETECTION	TOTAL
LATENCY	1 ms.	71 ms.	72 ms.
RAM	4.0K	612.6K	612.6K
FLASH	-	7.1M	-
ACCURACY			-

Unoptimized (float32)

Select

	IMAGE	OBJECT DETECTION	TOTAL
LATENCY	1 ms.	42 ms.	43 ms.
RAM	4.0K	509.7K	509.7K
FLASH	-	27.5M	-
ACCURACY			-

Run this model

Scan QR code or launch in browser to test your prototype



Launch in browser

?

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```
marc@pi4:~ $ edge-impulse-linux-runner
Edge Impulse Linux runner v1.18.2
```

```
[RUN] Already have model /home/marc/.ei-linux-runner/models/783798/v3-quantized-runner-linux-aarch64/model.eim not downloading...
[RUN] Starting the image classifier for Arduino Edge Impulse / Object Detection - Rubber Duckies (v3)
[RUN] Parameters image size 96x96 px (3 channels) classes [ 'rubber-ducky' ]
[RUN] Thresholds: 5.min_score=0.5 (override via --thresholds <value>)
[GST] checking for /etc/os-release
[RUN] Connected to camera Logitech BRIO (/dev/video0)
```

```
Want to see a feed of the camera and live classification in your browser? Go to http://192.168.1.136:4912
```

```
Want to use predictions in your application? Open a websocket to ws://192.168.1.136:4912
```

```
boundingBoxes 3ms. [{"height":8,"label":"rubber-ducky","value":0.9609375,"width":16,"x":16,"y":64}]
boundingBoxes 2ms. [{"height":8,"label":"rubber-ducky","value":0.96484375,"width":16,"x":16,"y":64}]
boundingBoxes 1ms. [{"height":8,"label":"rubber-ducky","value":0.98046875,"width":16,"x":16,"y":64}]
boundingBoxes 2ms. [{"height":8,"label":"rubber-ducky","value":0.96875,"width":16,"x":16,"y":64}, {"height":8,"label":"rubber-ducky","value":0.55859375,"width":8,"x":72,"y":80}]
boundingBoxes 1ms. [{"height":8,"label":"rubber-ducky","value":0.95703125,"width":16,"x":16,"y":64}, {"height":8,"label":"rubber-ducky","value":0.59765625,"width":8,"x":72,"y":80}]
boundingBoxes 1ms. [{"height":8,"label":"rubber-ducky","value":0.98046875,"width":16,"x":16,"y":64}, {"height":8,"label":"rubber-ducky","value":0.5390625,"width":8,"x":72,"y":80}]
boundingBoxes 2ms. [{"height":8,"label":"rubber-ducky","value":0.96875,"width":16,"x":16,"y":64}, {"height":8,"label":"rubber-ducky","value":0.51953125,"width":8,"x":72,"y":80}]
boundingBoxes 2ms. [{"height":8,"label":"rubber-ducky","value":0.98046875,"width":16,"x":16,"y":64}, {"height":8,"label":"rubber-ducky","value":0.55859375,"width":8,"x":72,"y":80}]
```

Object Detection - Rubber D...

Edge Impulse for Linux

No segur

192.168.1.136:4912

☆

🔖

☰

Feina

Finalitza l'actualització

Arduino Edge Impulse / Object Detection - Rubber Duckies

More exercises

- Is it better to use another model (YoloV5? YoloPRO?)?
 - Accuracy?
 - Performance?
- What happens if you re-train the model with the less represented ducks? (e.g. green, pink, white?)

My classes (4)

- ~~Business models and the Internet of Things.~~

- ~~Business Model Canvas~~

—

- ~~IoT / Edge AI project (hands on)~~

—

- The IoT Stack - How to build your IoT project, digital transformation and more.

- IoT Model Canvas

—

- Scale your IoT / Edge AI projects (hands-on)